Math 5601: Introduction to Numerical Analysis
Homework assignment 2

Show all relevant work in detail to justify your conclusions. Partial credit depends upon the work you show. For each numerical experiment, all the .m files of your Matlab code should be electronically submitted to qingguohong@mst.edu together with a .txt file which copies all the information in the Matlab command window when you run the code to obtain the numerical results.

Problem #1: Define a special function $f(x)$ for Newton's method such that Newton's method can lead to the following iterative scheme

$$x_{k+1} = x_k(2 - Rx_k)$$

to calculate the reciprocal of a number R. In order to fully justify your $f(x)$, you need to use the scheme of Newton's method to derive the above iterative scheme in details.

Problem #2: Program the Newton's method $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$ in Matlab. Consider $f(x) = arctan(x)$ which has a unique zero $z = 0$. Choose all the tolerances to be 10^{-10}. Starting with

$$x_0 = 0.5, 1, 1.3, 1.4, 1.35, 1.375, 1.3875, 1.39375, 1.390625, 1.3921875,$$

iterate until one of the tolerance is satisfied or 20 iterations are completed. Output the numerical solution at each iteration step. What behavior do you observe?

Problem #3: Program the secant method $x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$ in Matlab. Consider $f(x) = arctan(x)$ which has a unique zero $z = 0$. Choose all the tolerances to be 10^{-10}. Starting with

$$(x_0, x_1) = (0.5, 1), (1, 1.3), (1.4, 1.5), (10, 11),$$

iterate until one of the tolerance is satisfied or 20 iterations are completed. Output the numerical results at each iteration step.

Math 560 - Homework 02 - Tyler Glidden

Problem 1

Define a function $F(x)$ s.t. Newton's method converges to

$$x_{k+1} = x_k(2 - Rx_k)$$ to calculate the reciprocal of a number R

$F(x) = \frac{1}{x}$ gives a reciprocal, but has no zero's and is not helpful.

by shifting the graph up/down we can force it to have a zero.

try: $F(x) = \frac{1}{x} - R$.

$\Rightarrow F'(x) = \frac{-1}{x^2}$

$\Rightarrow g(x) = x - \frac{F(x)}{F'(x)}$ gives Newton's method

$$g(x) = x - \frac{\frac{1}{x} - R}{-1/x^2} = x + \left(\frac{1}{x} - R\right) \cdot x^2 = x + x - Rx^2 = 2x - Rx^2 = x(2 - Rx)$$

For a given R, the zero given will be the Reciprocal of R.

See Matlab example.

Problem 2

Program Newton's Method for $f(x) = \arctan(x)$

- all $\varepsilon = 10^{-10}$, $M = 20$

- $x_0 = 1.4 \rightarrow$ Diverge
- $x_0 = 1.39375 \rightarrow$ Diverge
- $x_0 = 1.3921875 \rightarrow$ Diverge

Convergence Theory:

Since this is a fixed point method,

$|g'(x)| \leq L$ means we have convergence.

↳ choose $L = 1$ for convergence

$$g(x) = x - \frac{f(x)}{f'(x)} \qquad g'(x) = 1 - \frac{f'(x)\cdot f'(x) - f(x)\cdot f''(x)}{(f'(x))^2} = \frac{(f'(x))^2 - (f'(x))^2 - f(x)\, f''(x)}{(f'(x))^2}$$

$f(x) = \arctan x$

$f'(x) = \frac{1}{1+x^2}$

$f''(x) = \frac{-2x}{(1+x^2)^2}$

$$= -\frac{f(x)\cdot f''(x)}{(f'(x))^2} = \frac{\arctan(x)\cdot -2x}{\frac{1\cdot(1+x^2)}{(1+x^2)^2}} = -2x\arctan(x)$$

$$|-2x\arctan(x)| \leq 1$$

$$-1 \leq -2x\arctan(x) \leq 1$$

$-1 \leq -2x\arctan(x)$

$0 \leq -2x\arctan(x) + 1$

solve

$0 = -2x\arctan(x) + 1$

$x \approx 0.7654$ //

↳ only solved numerically with bisection

$-2x\arctan(x) \leq 1$

$-2x\arctan(x) - 1 \leq 0$

solve

$-2x\arctan(x) - 1 = 0$

$x \approx$ All Real

or (easier, but less accurate)

$$-2x\arctan(x) \leq |-2x\,x| \leq 1$$

$|-2x^2| \leq 1$

$-1 \leq -2x^2 \leq 1$

$+1 \geq +2x^2$

$\frac{1}{2} \geqslant x^2$

$x \leq \frac{1}{\sqrt{2}}$ //

$x \leq 0.7071$ //

$-x^2 \leq \frac{1}{2}$

for all real

Conclusion:

Based on the convergance theory, the interval from $[-0.7654, 0.7654]$ or $[-1/\sqrt{2}, 1/\sqrt{2}]$ guarentees convergence. However, there exists an interval closer to $[-1.390625, 1.390625]$ s.t. the newton's method converges.

Problem 3

Program Secant method for $f(x) = \arctan(x)$

all $\varepsilon = 10^{-10}$, $M = 20$

$(x_0, x_1) = (0.5, 1), (1, 1.3), (1.4, 1.5), (10, 11)$